GAMMA-POSITIVITY OF h-POLYNOMIALS OF PREORDERS

KNOWLEDGE SENSE

Abstract. For a preorder τ on an n-element set, Athanasiadis and Chapoton defined the h-polynomial $h(\tau,t)$, which counts the lattice points of the preorder polytope Q_τ by their number of nonzero coordinates, and conjectured that it is γ-positive. We prove this conjecture for every finite preorder. More precisely, we show that

$$h(\tau,t)=\sum_{j=0}^{\lfloor n/2\rfloor} b_j(\tau)\,t^j(1+t)^{n-2j},$$

where $b_j(\tau)$ is the number of ordered pairs (I,J) of disjoint j-element subsets for which there is a bijection $f\colon I\to J$ with $i\le f(i)$ for all i. The proof combines a support duality for draconian sequences of bipartite graphs, due to Dai, Hou, Liu, Thawinrak and Wang, with a short argument that reduces arbitrary pairs of subsets admitting such a bijection to disjoint ones. As special cases we recover the known γ-expansions for total orders, ordinal sums of two antichains, octopuses and lopsided octopuses.

1. Introduction

Let E be a finite set with n elements. A *preorder* on E is a reflexive and transitive relation $\le$ on E. Antisymmetry is not assumed, so $\le$ may be viewed as a partial order on the blocks of a partition of E. A subset $U\subseteq E$ is an *order ideal* if $y\in U$ and $x\le y$ imply $x\in U$. Following Athanasiadis and Chapoton [3], the *preorder polytope* of $\tau=(E,\le)$ is

$$Q_\tau=\Big\{x\in\mathbb{R}^E_{\ge 0}\ :\ \sum_{e\in U}x_e\le |U| \text{ for every order ideal } U \text{ of } \tau\Big\},$$

and the *h-polynomial* of τ is

$$h(\tau,t)=\sum_{x\in Q_\tau\cap\mathbb{Z}^E} t^{|\operatorname{supp}(x)|},\qquad \operatorname{supp}(x)=\{e\in E: x_e\neq 0\}.$$

When τ comes from an arbor, Q_τ is the arbor polytope of Chapoton [5], and $h(\tau,t)$ is the polynomial introduced there. Note that $h(\tau,t)$ is not the h^*-polynomial of Q_τ.

A polynomial $h(t)$ is *γ-positive with center of symmetry $n/2$* if $h(t)=\sum_{j=0}^{\lfloor n/2\rfloor}\gamma_j t^j(1+t)^{n-2j}$ with $\gamma_j\ge 0$. Since the polynomials $t^j(1+t)^{n-2j}$ are linearly independent, the coefficients γ_j are uniquely determined. The γ-positivity of $h(t)$ implies that its coefficient sequence is palindromic and unimodal; see the survey [1]. Athanasiadis and Chapoton [3, Conjecture 5.2] conjectured that $h(\tau,t)$ is γ-positive for every preorder τ; for arbors this was implicit in [5]. They verified the conjecture by computer for all preorders of size at most 7. Special cases were settled earlier: chain preorders with blocks of arbitrary sizes in [2], ordinal sums of two antichains and zig-zag posets in [3, Section 9] (the latter using [11]), the arbors called octopuses in [4] (with a bijective proof in [8]), and lopsided octopuses in [8].

Date: October 7, 2026.

2020 *Mathematics Subject Classification.* 05A15, 52B20, 06A07.

Key words and phrases. Preorder, preorder polytope, lattice point, γ-positivity, draconian sequence, perfect matching.

In this paper we prove the conjecture in general, with an explicit combinatorial description of the γ-coefficients.

Definition 1.1. Let $I, J \subseteq E$. An *upward bijection* from I to J is a bijection $f\colon I \to J$ such that $i \le f(i)$ for every $i \in I$. For $0 \le j \le \lfloor n/2 \rfloor$, let $b_j(\tau)$ be the number of ordered pairs (I, J) of *disjoint* j-element subsets of E for which an upward bijection $I \to J$ exists.

Here i and $f(i)$ may lie in the same block of τ. Each pair (I, J) is counted once, even if several upward bijections $I \to J$ exist. The empty pair gives $b_0(\tau) = 1$.

Theorem 1.2. *For every preorder τ on an n-element set,*

$$h(\tau, t) = \sum_{j=0}^{\lfloor n/2 \rfloor} b_j(\tau)\, t^j (1+t)^{n-2j}. \tag{1.1}$$

In particular, $h(\tau, t)$ is γ-positive with $\gamma_j(\tau) = b_j(\tau)$, and Conjecture 5.2 of [3] holds.

Theorem 1.2 implies Conjecture 5.1(b) of [3], the palindromicity and unimodality of $h(\tau, t)$ for every preorder τ. Since arbors are special preorders, it gives in particular the palindromicity and unimodality of $h(\sigma, t)$ for every arbor σ, which Chapoton [5] conjectured as part of a stronger statement.

Corollary 1.3. *Let τ be a preorder of size n and let τ^* be its dual preorder.*

(a) *The coefficient sequence of $h(\tau, t)$ is palindromic and unimodal.*
(b) *$b_j(\tau) = b_j(\tau^*)$ for all j; hence $h(\tau, t) = h(\tau^*, t)$.*
(c) *$b_1(\tau)$ is the number of ordered pairs (x, y) of elements of E with $x \ne y$ and $x \le y$.*

Proof. Part (a) is a standard consequence of γ-positivity. For (b), if $f\colon I \to J$ is an upward bijection for τ, then $f^{-1}\colon J \to I$ is an upward bijection for τ^*, so $(I, J) \mapsto (J, I)$ is a bijection between the pairs counted by $b_j(\tau)$ and by $b_j(\tau^*)$. Part (c) is the case $j = 1$ of Definition 1.1. □

The identity $h(\tau, t) = h(\tau^*, t)$ is Conjecture 5.4 of [3], proved by Dai, Hou, Liu, Thawinrak and Wang [6]. Since our argument uses their duality theorem, Corollary 1.3(b) is not an independent proof.

Relation to previous work. Apart from Hall's theorem, the only result we use is the support duality of Dai, Hou, Liu, Thawinrak and Wang [6, Theorem 1.3], recalled as Theorem 2.1. From it we derive Lemma 2.2, which states that the number of draconian sequences with full support equals the number of vertex subsets admitting a perfect matching. Statements of this type are known; for instance, Ohsugi and Tsuchiya [9, Proposition 3.4] describe interior polynomials in terms of vertex sets of matchings (see Remark 7.1). The new ingredient is the passage from all pairs (I, J) admitting an upward bijection to disjoint ones (Lemma 3.2), which yields the expansion (1.1).

The paper is organized as follows. Section 2 recalls the duality theorem and derives the counting lemma. Section 3 proves Theorem 1.2, and Section 4 gives a second derivation. Section 5 treats examples, Section 6 reports a computational check, and Section 7 contains concluding remarks.

2. Draconian sequences and a counting lemma

Let H be a finite bipartite graph with vertex classes L and R. For $A \subseteq L$ write $N_H(A) \subseteq R$ for the set of neighbors of A. Following [6, Definition 1.1], which is a variant of Postnikov's

notion [10], let

$$\mathcal{D}(H) = \Big\{ a \in \mathbb{Z}_{\geq 0}^{L} \ : \ \sum_{i \in A} a_i \leq |N_H(A)| \text{ for every } A \subseteq L \Big\}$$

be the set of *H-draconian sequences*, and let $F_H(s) = \sum_{a \in \mathcal{D}(H)} s^{|\operatorname{supp}(a)|}$ be its support enumerator. In contrast with Postnikov's original definition, the total sum $\sum_i a_i$ is not required to take a fixed value; it is only bounded above by $|N_H(L)|$. Let H^* denote the same graph with the roles of L and R interchanged.

Theorem 2.1 ([6, Theorem 1.3])**.** *For every finite bipartite graph H, one has $F_H(s) = F_{H^*}(s)$.*

The graphs to which we apply Theorem 2.1 may have isolated vertices or an empty vertex class. These cases reduce directly to graphs with neither. Indeed, if v is an isolated vertex of H in L, then every $a \in \mathcal{D}(H)$ satisfies $a_v \leq |N_H(\{v\})| = 0$, while in H^* the vertex v lies in no neighborhood; so deleting v changes neither F_H nor F_{H^*}. The same holds for an isolated vertex in R, by symmetry. After all isolated vertices are deleted, either both classes are nonempty, or the graph has no vertices, in which case $F_H = F_{H^*} = 1$.

Lemma 2.2. *Let H be a bipartite graph with vertex classes L and R, and let $k = |L|$. Then*

$$\#\{a \in \mathcal{D}(H) : \operatorname{supp}(a) = L\} = \#\{I \subseteq R : |I| = k \text{ and } H[L \sqcup I] \text{ has a perfect matching}\}.$$

Proof. The left-hand side is the coefficient of s^k in $F_H(s)$, because every $a \in \mathcal{D}(H)$ has support contained in L. By Theorem 2.1 it equals the number of $b \in \mathcal{D}(H^*)$ with exactly k positive entries. Taking $A = R$ in the definition of $\mathcal{D}(H^*)$ gives $\sum_{r \in R} b_r \leq |N_{H^*}(R)| \leq k$, so such a vector b is the indicator vector $\mathbf{1}_I$ of a k-element subset $I \subseteq R$. It remains to decide when $\mathbf{1}_I \in \mathcal{D}(H^*)$. If $|B| \leq |N_{H^*}(B)|$ for every $B \subseteq I$, then for every $A \subseteq R$ we have $|A \cap I| \leq |N_{H^*}(A \cap I)| \leq |N_{H^*}(A)|$. Hence $\mathbf{1}_I \in \mathcal{D}(H^*)$ if and only if $|B| \leq |N_{H^*}(B)|$ for all $B \subseteq I$, and since $|I| = |L|$, by Hall's theorem [7] this holds if and only if $H[L \sqcup I]$ has a perfect matching. □

3. Proof of Theorem 1.2

Fix a preorder $\tau = (E, \leq)$. For $Y \subseteq E$ let $\downarrow Y = \{e \in E : e \leq y \text{ for some } y \in Y\}$. By reflexivity and transitivity, $\downarrow Y$ is the smallest order ideal containing Y.

For $J \subseteq E$, let H_J be the bipartite graph with vertex classes J and a disjoint copy of E, in which $j \in J$ is adjacent to $i \in E$ if and only if $i \leq j$. Thus $N_{H_J}(Y) = \downarrow Y$ for every $Y \subseteq J$.

Lemma 3.1. *Let $J \subseteq E$ and let $x \in \mathbb{Z}_{\geq 0}^{E}$ with $\operatorname{supp}(x) = J$. Then $x \in Q_\tau$ if and only if the restriction $x|_J$ belongs to $\mathcal{D}(H_J)$.*

Proof. Suppose $x \in Q_\tau$ and let $Y \subseteq J$. Since x is nonnegative and vanishes off J, and $\downarrow Y$ is an order ideal containing Y,

$$\sum_{j \in Y} x_j \leq \sum_{e \in \downarrow Y} x_e \leq |\downarrow Y| = |N_{H_J}(Y)|.$$

Conversely, suppose $x|_J \in \mathcal{D}(H_J)$ and let U be an order ideal. Put $Y = J \cap U$. Then $N_{H_J}(Y) = \downarrow Y \subseteq U$, hence $\sum_{e \in U} x_e = \sum_{j \in Y} x_j \leq |N_{H_J}(Y)| \leq |U|$. □

By Lemma 3.1, the number of lattice points of Q_τ with support exactly J equals the number of elements of $\mathcal{D}(H_J)$ with full support J. By Lemma 2.2, this is the number of subsets $I \subseteq E$

with $|I| = |J|$ for which $H_J[J \sqcup I]$ has a perfect matching. A perfect matching of $H_J[J \sqcup I]$ is the same as an upward bijection $I \to J$. Summing over J we obtain

$$h(\tau, t) = \sum_{(I,J)} t^{|J|}, \tag{3.1}$$

where the sum runs over all ordered pairs (I, J) of subsets of E, *not necessarily disjoint*, for which an upward bijection $I \to J$ exists. Each such pair is counted once.

It remains to remove the elements common to I and J. For a pair (I, J) put

$$A = I \setminus J, \qquad B = J \setminus I, \qquad K = I \cap J.$$

Lemma 3.2. *There is an upward bijection $I \to J$ if and only if there is an upward bijection $A \to B$.*

Proof. Suppose $f\colon I \to J$ is an upward bijection, and consider the directed graph on $I \cup J$ with arrows $i \to f(i)$. Each vertex of A has out-degree 1 and in-degree 0, each vertex of B has in-degree 1 and out-degree 0, and each vertex of K has in-degree and out-degree 1. Hence the connected components are directed cycles contained in K (including loops $f(i) = i$) and directed paths that start in A, pass through K and end in B. Every element of A starts exactly one path and every element of B ends exactly one path. Sending the initial vertex of each path to its terminal vertex defines a bijection $g\colon A \to B$, and transitivity applied along each path gives $a \le g(a)$.

Conversely, if $g\colon A \to B$ is an upward bijection, then g together with the identity map of K is an upward bijection $I \to J$, by reflexivity. □

Proof of Theorem 1.2. The map $(I, J) \mapsto (A, B, K)$ is a bijection from ordered pairs of subsets of E with $|I| = |J|$ to triples of pairwise disjoint subsets with $|A| = |B|$; its inverse is $(A, B, K) \mapsto (A \cup K, B \cup K)$, and $|J| = |B| + |K|$. By Lemma 3.2, the pairs occurring in (3.1) correspond to the triples in which (A, B) is one of the disjoint pairs counted by $b_j(\tau)$, where $j = |A| \le \lfloor n/2 \rfloor$, and K is an arbitrary subset of the $(n-2j)$-element set $E \setminus (A \cup B)$. Therefore

$$h(\tau, t) = \sum_{j=0}^{\lfloor n/2 \rfloor} \sum_{(A,B)} \sum_{K \subseteq E \setminus (A \cup B)} t^{\,j+|K|} = \sum_{j=0}^{\lfloor n/2 \rfloor} b_j(\tau)\, t^j (1+t)^{n-2j},$$

where (A, B) runs over the pairs counted by $b_j(\tau)$. □

Remark 3.3. The preorder axioms enter the proof in three places: reflexivity and transitivity make $\downarrow Y$ the smallest order ideal containing Y (Lemma 3.1), transitivity shortens paths in Lemma 3.2, and reflexivity extends a bijection $A \to B$ by the identity on K.

4. A second derivation

We sketch an equivalent derivation, which produces the disjoint pairs directly. For $x \in \mathbb{Z}^E_{\ge 0}$ put

$$Z = \{e : x_e = 0\}, \qquad J = \{e : x_e \ge 2\}, \qquad d_a = x_a - 1 \quad (a \in J),$$

so that d_a is the excess of x_a over 1. For every $U \subseteq E$ one has $\sum_{e \in U} x_e - |U| = \sum_{a \in J \cap U} d_a - |Z \cap U|$. Let $H_{J,Z}$ be the bipartite graph with vertex classes J and Z, in which $a \in J$ is adjacent to $z \in Z$ when $z \le a$. Arguing as in Lemma 3.1 (apply the defining inequality to $\downarrow Y$ for $Y \subseteq J$, and for an order ideal U use $Y = J \cap U$ and $N_{H_{J,Z}}(Y) \subseteq Z \cap U$), one finds that $x \in Q_\tau$ if and only if $d \in \mathcal{D}(H_{J,Z})$; note that d has full support J. By Lemma 2.2, for fixed disjoint Z and J the number of such x equals the number of j-element subsets $I \subseteq Z$, $j = |J|$,

admitting an upward bijection $I \to J$. Since $|\operatorname{supp}(x)| = n - |Z|$, reordering the finite sum gives

$$h(\tau, t) = \sum_{(I,J)} \sum_{I \subseteq Z \subseteq E \setminus J} t^{\,n-|Z|} = \sum_{j=0}^{\lfloor n/2 \rfloor} b_j(\tau)\, t^j (1+t)^{n-2j},$$

where (I, J) runs over the disjoint pairs counted by the numbers $b_j(\tau)$. Here the coordinates equal to 1 play the role of the set K in Section 3.

5. Examples

In each of the following cases $b_j(\tau)$ is computed directly from Definition 1.1.

Example 5.1 (Antichains and a single block)**.** If τ is an antichain, the only upward bijections are identity maps, so $b_j(\tau) = 0$ for $j \geq 1$ and $h(\tau, t) = (1+t)^n$. If τ has a single block of size n, every pair of disjoint subsets of equal size qualifies, and $b_j(\tau) = \binom{n}{j}\binom{n-j}{j}$.

Example 5.2 (Total orders)**.** Let τ be the total order $1 < 2 < \cdots < n$, that is, the chain preorder all of whose blocks are singletons. For disjoint $I = \{u_1 < \cdots < u_j\}$ and $J = \{v_1 < \cdots < v_j\}$, an upward bijection exists if and only if $u_k < v_k$ for all k. Choosing the $2j$ elements of $I \cup J$ and then a ballot sequence gives

$$b_j(\tau) = \binom{n}{2j} C_j, \qquad C_j = \frac{1}{j+1}\binom{2j}{j}.$$

This is the classical γ-expansion of the Narayana polynomial (see [1]), in agreement with [5], where $h(\tau, t)$ is identified with the h-polynomial of the associahedron of type A_n. Chain preorders with larger blocks are covered by Theorem 1.2 and by [2]; we do not compute their coefficients here.

Example 5.3 (Ordinal sum of two antichains)**.** Let τ be the ordinal sum of an antichain of size p below an antichain of size q. An element of the top level lies below no element other than itself, and distinct elements of the bottom level are incomparable. Hence a disjoint pair (I, J) admits an upward bijection exactly when I lies in the bottom level and J in the top level, and $b_j(\tau) = \binom{p}{j}\binom{q}{j}$. This recovers [3, Proposition 9.2].

Example 5.4 (Octopuses and lopsided octopuses)**.** Let $n, k_1, k_2 \geq 0$ with $k = k_1 + 2k_2 < n$, and let τ be the lopsided octopus of [8]: a root block R of size $n-k$, together with k_1 singleton leaves and k_2 two-element leaves, each leaf lying directly below R. An element of R lies below only elements of R, and an element of a leaf lies below only R and the elements of its own leaf. Consider a disjoint pair (I, J) with an upward bijection. No element of J lies in a singleton leaf, and an element of J lying in a two-element leaf must be matched with the other element of that leaf; all other elements of J lie in R, and they can be matched with arbitrary remaining elements. If exactly i two-element leaves meet J, there are $2^i\binom{k_2}{i}$ choices for these leaves and for the element of each of them lying in J, then $\binom{n-k}{j-i}$ choices for $J \cap R$, and $\binom{n-j-i}{j-i}$ choices for the remaining $j-i$ elements of I, which are taken outside J and outside the i chosen leaves. Thus

$$b_j(\tau) = \sum_{i=0}^{k_2} 2^i \binom{k_2}{i}\binom{n-k}{j-i}\binom{n-j-i}{j-i},$$

which is [8, Proposition 3.4]. For $k_2 = 0$ this is $\binom{n-k}{j}\binom{n-j}{j}$, the formula of Athanasiadis, Xiao and Yan [4, Proposition 2.2] for octopuses.

Example 5.5 (A poset that is not an arbor)**.** Let $E = \{a_1, a_2, a_3, d, c_0, c_1, c_2, c_3\}$ with the strict relations $a_i < c_0$, $a_i < c_i$ and $d < c_i$ for $i = 1, 2, 3$; see Figure 1. Then

$$h(\tau, t) = 1 + 17t + 106t^2 + 303t^3 + 427t^4 + 303t^5 + 106t^6 + 17t^7 + t^8,$$

so that $(b_0(\tau), \ldots, b_4(\tau)) = (1, 9, 24, 16, 1)$. Here $b_1(\tau) = 9$ is the number of ordered pairs $x \neq y$ with $x \leq y$, which are the strict relations since τ is a poset; this is in agreement with Corollary 1.3(c). Also $b_4(\tau) = 1$, because $(\{a_1, a_2, a_3, d\}, \{c_0, c_1, c_2, c_3\})$ is the only disjoint pair of 4-element sets with an upward bijection.

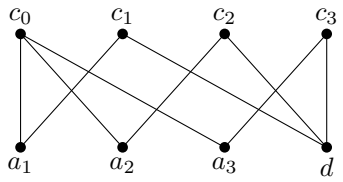


FIGURE 1. The Hasse diagram of the poset of Example 5.5.

6. Computational check

An exact enumeration in integer arithmetic confirms (1.1) for all 7,332 labeled preorders on at most five elements, a count that includes the empty preorder, for 100 pseudorandom preorders on six to eight elements, and for Example 5.5. It also confirms Theorem 2.1 and Lemma 2.2 for all 689 labeled bipartite graphs with at most three vertices in each class. The program enumerates lattice points of Q_τ directly from the order-ideal inequalities and computes $b_j(\tau)$ by testing for the existence of an upward bijection. These computations are not part of the proof. The program is available from the author.

7. Concluding remarks

Remark 7.1 (Interior polynomials)*.* Let G_τ be the bipartite graph whose vertex classes are two copies of E, with i in the first copy adjacent to j in the second whenever $i \leq j$. The pairs (I, J) in (3.1) are exactly the pairs of vertex sets of matchings of G_τ, so (3.1) is a generating function of the type that Ohsugi and Tsuchiya [9, Proposition 3.4] use to describe interior polynomials. In [9, Theorem 0.3], numbers of such vertex sets, multiplied by powers of 4, appear as γ-coefficients of the h^*-polynomial of a different reflexive polytope $\mathcal{B}_G$; that result concerns a different polynomial from $h(\tau, t)$.

Remark 7.2 (Scope)*.* Theorem 1.2 implies Conjectures 5.1(a), 5.1(b) and 5.2 of [3]. It does not address the realization of $h(\tau, t)$ as the h-polynomial of a (flag) simplicial polytope [3, Conjecture 5.1(c),(d)], the real-rootedness of $h(\tau, t)$ [3, Conjecture 5.3], or the conjectures of [3] on the h^*-polynomial of Q_τ. Since $h(\tau, t)$ is γ-positive, its real-rootedness is equivalent to that of the polynomial $\sum_j b_j(\tau) t^j$.

Acknowledgments

AI-assisted tools were used in the research and preparation of this manuscript.

References

[1] C. A. Athanasiadis, *Gamma-positivity in combinatorics and geometry*, Sém. Lothar. Combin. **77** (2018), Article B77i.

[2] C. A. Athanasiadis, *Lattice point enumeration of polytopes associated to integer compositions*, Ann. Comb. (2026), doi:10.1007/s00026-026-00812-2.

[3] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1 (2026).

[4] C. A. Athanasiadis, Q. Xiao and X. Yan, *Lattice point enumeration of some arbor polytopes*, arXiv:2603.11654 (2026).

[5] F. Chapoton, *On posets and polytopes attached to arbors*, Math. Scand. **131** (2025), 401–449.

[6] Z. Dai, Q. Hou, Z. Liu, W. Thawinrak and H. Wang, *Counting lattice points in Minkowski sums of cross polytopes*, arXiv:2608.16037v2 (2026).

[7] P. Hall, *On representatives of subsets*, J. London Math. Soc. **10** (1935), 26–30.

[8] K. Menon, *Gamma-positivity for octopuses: a bijective proof*, arXiv:2608.13247v1 (2026).

[9] H. Ohsugi and A. Tsuchiya, *Reflexive polytopes arising from bipartite graphs with γ-positivity associated to interior polynomials*, Selecta Math. (N.S.) **26** (2020), Article 59, doi:10.1007/s00029-020-00588-0.

[10] A. Postnikov, *Permutohedra, associahedra, and beyond*, Int. Math. Res. Not. IMRN **2009** (2009), 1026–1106.

[11] Y. Wang, S.-N. Zheng and X. Chen, *Analytic aspects of Delannoy numbers*, Discrete Math. **342** (2019), 2270–2277.

Knowledge Sense

Email address: `math@knowledgesense.jp`